

Haematuria and urology red flags

1. Visible haematuria

(Based on NICE 2015, NG12, BMJ 2022;376:e067395, BMJ 2009;338:a3021, investigation of adult haematuria in Oxfordshire in primary and secondary care, 2008 and pragmatism)



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Specific recommendations from the NICE guidance on suspected cancer are marked as NG12 throughout

CHILD with unexplained visible haematuria

VERY URGENT referral (≤48h) (NG12)

Call and discuss with paed's ?Wilms' tumour

ADULT with visible haematuria

(Do NOT attribute to antiplatelet/anticoagulant: investigate!)

Any symptoms/signs of a UTI?

YES

NO

Treat as UTI, send culture and ensure follow-up with urine dip 7–14 days after antibiotics

Follow-up:

- Did culture (if done) confirm a UTI?
- Have symptoms settled?
- Has haematuria resolved after treatment? Check dipstick.

NO to ANY

YES to ALL

Reassuring

Safety-net: reconsult if visible haematuria recurs

INVESTIGATE as per NG12:

All unexplained visible haematuria
or
Visible haematuria that persists/recurs after successful treatment of a UTI

Age ≥45y

Age <45y

In men, also consider a DRE and PSA (NG12)

Suspected cancer pathway referral (NG12)

Investigate as appropriate
Follow local pathway

Cause found?

NO

YES

Consider renal/urology referral

In women ≥55y, consider transvaginal USS (NG12) to assess endometrial thickness if:

- Visible haematuria and ANY ONE of:
 - Thrombocytosis
 - Anaemia
 - High blood glucose
 - Unexplained vaginal discharge

Manage as appropriate

Causes of visible and non-visible haematuria include:

- Cancer: bladder, kidney, prostate, endometrial
- UTI
- Renal causes: e.g. stone, glomerulonephritis, IgA nephropathy, polycystic kidney disease: bleed into a cyst
- Benign prostatic disease (e.g. BPH, prostatitis)
- Trauma or exercise-induced haematuria
- Rarely: renal infarction, TB
- Uncontrolled anticoagulation
- Red-coloured urine: rhabdomyolysis, drugs (e.g. rifampicin), foods (e.g. beetroot/ blackberries)

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2. Non-visible haematuria

Based on NICE 2015, NG12, BJGP2014;DOI:10.3399/bjgp14X681409, BMJ 2009;338:a3021 and pragmatism.



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Non-visible haematuria: pitfalls

- Non-visible haematuria is common in those WITHOUT cancer: affects 2.5–20% of the population.
- Causes of haematuria are detailed at the bottom of page 1.
- The positive predictive value (PPV) (the chance of someone with the symptom having cancer) of non-visible haematuria in someone ≥ 60 y without other symptoms is 1.6%.
- Non-visible haematuria in someone ≥ 60 y combined with either dysuria or a raised white cell count has a PPV $\geq 3\%$ (NICE threshold for investigation is $\geq 3\%$).

Do not attribute non-visible haematuria to antiplatelet or anticoagulant therapy – investigate!

Is the non-visible haematuria SIGNIFICANT or not?

NOT SIGNIFICANT

- Trace of blood on dipstick.
- Non-visible haematuria with a UTI which resolves on treatment (repeat urine dipstick after treatment).

Caution: store dipsticks in original container with lid ON and check expiry date before use: prolonged exposure to air may give a false positive result.

SIGNIFICANT

- Any single episode $\geq 1+$ of blood on dipstick and **symptoms** in the absence of UTI (or that persists after treatment).
- Persistent $\geq 1+$ blood on 2 out of 3 dipsticks, on different days, in an **asymptomatic** person.

Yes

≥ 60 y with unexplained non-visible haematuria **AND either:**
dysuria or raised WCC (check FBC)

NO

YES

Is it asymptomatic or symptomatic?

Suspected cancer pathway referral (NG12)

Asymptomatic unexplained non-visible haematuria
(Present on 2 out of 3 dipsticks on different days)

Symptomatic unexplained non-visible haematuria

Check BP, renal function and ACR/PCR

At Red Whale, we would also suggest considering a renal USS.

<40 y

≥ 40 y

ALL of:

- eGFR ≥ 60 ml/min.
- ACR <30 or PCR <50 .
- BP $<140/90$.

ANY of:

- eGFR <60 ml/min.
- ACR ≥ 30 or PCR ≥ 50 .
- BP $\geq 140/90$.

Nephrology referral

REFER

even if investigations normal

Urology referral

Primary care monitoring

Annual: BP, renal function, ACR/PCR for as long as non-visible haematuria persists.

Review if symptoms change/develops visible haematuria or if BP, renal function or ACR/PCR abnormal.

At Red Whale, we feel there are times we may contact secondary care for patients in this subgroup, e.g. a young patient with an eGFR in the 60s or if eGFR was falling. Use clinical judgement/contact secondary care if concerned.

3. Urinary symptoms: red flags and referral

Based on NICE 2015, NG12, PHE: Diagnosis of urinary tract infections, quick reference tool for primary care 2019, NICE CKS 2020, BMJ 2013;346:f3140 and pragmatism.



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Recurrent urinary symptoms: consider red flags

WOMEN

If increased urgency/frequency, (persistent or frequent) especially if >50y (NG12)

Check CA125
(urgent USS if CA125 raised)
(NG12)

MEN or WOMEN

If ≥60y and unexplained recurrent/persistent UTI (NG12)

Consider non-urgent referral
(?bladder cancer)
(NG12)

MEN

Any lower urinary tract symptoms (e.g. nocturia, frequency, hesitancy, urgency or retention) or erectile dysfunction (NG12)

Consider PSA and DRE
(NG12)

BUT FIRST: is it really an infection? And how do we pick these people up in practice?

- A recurrent UTI is defined as ≥2 infections in 6m or ≥3 infections in 1y.
- (Return of symptoms within 2w of an infection is likely a relapse rather than a recurrent UTI).
- Urine dipstick for infection can be unreliable in women ≥65y so we may not have done a dipstick so far, and, in a woman <65y, we may not have sent urine for culture.
- For a patient ≥60y with recurrent/persistent UTI, culture and dipstick are useful to aid investigation.

Practical application in clinical practice

Patient ≥60y presents with symptoms of a UTI...again!

Filing results: urine sample in a person ≥60y
Why was it done?

Review notes: look for recurrent UTIs or symptoms that don't resolve with treatment.

- Dipstick urine for blood.
- Send urine for culture.

If ≥1+ non-visible haematuria:
follow non-visible haematuria
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- Request a further urine sample:
- Dipstick urine for blood.
 - Check culture result.

Urine culture result

Urine shows infection

Consider non-urgent referral if unexplained recurrent/persistent infection as above (NG12)

Urine DOES NOT show infection

Check CA125 in women or consider PSA/DRE in men as above (NG12)

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